IKT200-Assignment 4- Solution

Task 1: Thinking about prototyping

Answer the following questions (Textbook Chapter 4)

a) Why should we do prototyping before production?
By creating a prototype, you get a good overview of the product. If a problem arises, it is cheaper and easier to make changes in the prototype than in a series of finished products.

b) What is meant by "physical sketches"?
Physical sketches are a step in prototyping where a (non-functioning) physical model is built. Here you can use Lego, on app, or similar.

c) Why do we make mockups?
Visual mockups can be detailed so that you end up with a good overview structure, content and layout. Creating mockups saves time and effort later in the process because changes to mockups are much easier than changes to the product/code. Engineering expression: "You can fix it now, on the drawing board, with a whisper - or later on the construction site, with a sledgehammer."

d) Give three examples of how we can create mockups using computer programs. (Give at least 3)
Photoshop, wireframe, interactive pdf, interactive wireframe (Balsamiq) , tool for app / website mockup (InVision), html / css / similar.

e) What is the difference between use cases and user stories?

* **Use cases** can be used to describe what users (actors) are doing to the system. It can give us a better understanding of how the system should be designed. Can also help those involved in getting the same idea of what the system should achieve and contain.

* **User stories** are short, simple descriptions of goals from the perspective of the user. "As a <type of user>, I want <some goal> so that <some reason>" Example: "As a <rest a uranium customer>, I want to <order the food via my smartphone> so that <I do not have to wait for waiter>.

f) Give examples of how cost vs. efficiency can affect prototyping, e.g. for selecting electronics components for our system.

* A microcontroller is a cheap way to build electronics but requires you to figure out where to connect wires, etc.

* A slightly more expensive, but simpler (and more efficient) variant is an **Arduino**. Same microcontroller, but you get it done on a card with marking of connection points, etc. *(Very good solution in many cases!)*

* Similarly, you can invest in a more expensive hardware solution that already has Linux software or similar installed. (For example: **Raspberry Pi**)

* An expensive variant: Prototype on a **smartphone or PC** (if you have everything suitable phone / PC it will not be expensive).

g) What is the basic conflict of interest between open source and closed source? Also describe the advantages and disadvantages of each variant.

In its simplest form, we can say that closed source vs. Open source is about two things: 1) Your rights to the code you created (your "Intellectual property"). 2) Other people's rights to freely

modify and reuse your code. Often, you see benefits on both sides: * As developers, we want the opportunity to look at, learn from, and modify existing code and solutions. * At the same time, we want to protect what we make ourselves.

Task 2: Pathfinding, intro

What three parts does pathfinding consist of? Give a brief description of each, in your own words.

1) **Path node definition** (search points, predefined): Must have defined points for the machine to move, usually in the form of grid, hexgrid or you can customize points yourself.

2) **Path searching** (finding the best path to the goal): This can be controlled with static path searching (table lookup) or dynamically controlled with search algorithms, such as Dijkstra or A *.

3) **Path following**: To perform the move from start to finish.